

POS System Development Report (Post May 24, 2026)

1. Retail POS Migration & Enhancements (formerly Grocery POS)

- **Schema & Rebranding:** Renamed "Grocery" plans, features, and database enums to "Retail" globally to support broader business types (e.g. pharmacies, supermarkets).
- **Security Check:** Enforced role filtering in the UI so that Retail companies don't see restaurant roles like Chef or Waiter.
- **Seeding Logic:** Updated backend database seeding scripts (`seed-retail-plans.ts`) to automatically seed "Retail Starter" and "Retail Pro" plans.
- **POS Interface Enhancements:** Improved the Retail POS page with a robust floating cart, fast search bar, category filtering, and barcode scanning integration.
- **Receipts & Partial Payments:** Fixed the POS receipt to accurately display "Due Amount" when customers make partial payments (instead of confusingly trying to calculate change).
- **TypeScript & Build Fixes:** Resolved all lingering TypeScript compilation errors during the build process regarding the `auth.service.ts` logic.

2. Inventory & Purchase Returns Module

- **Purchase Returns:** Built a complete module for returning damaged/expired products to suppliers (pending -> approved -> settled lifecycle).
- **Barcode & SKU Support:** Added native Barcode and SKU inputs to product forms and enabled `findByBarcode` lookups for fast checkout scanning via USB or camera.

3. Subscription & Landing Page Overhaul

- **Dynamic Pricing Table:** Built a clean toggle on the landing page so visitors can switch between "Restaurant" and "Retail" plans.
- **Registration Filtering:** Connected the signup flow so that users only see the plans relevant to their selected business type.
- **Auto-Seeding:** Implemented lifecycle hooks to ensure subscription plans are automatically inserted on application startup.

4. Security & Hardening Fixes

- **NoSQL Injection Guard:** Added global `ObjectIdValidationMiddleware` to stop malformed requests from crashing the server.
- **Environment Fallbacks:** Removed dangerous hardcoded keys (JWT, Mongo URI) forcing the system to rely strictly on secure environment variables.
- **Log Sanitization:** Cleaned up the authentication service to prevent any sensitive data (like passwords) from leaking in server logs.
- **Request Limiting:** Hardened CORS policies and tightened Express payload limits to 5MB.

5. UI/UX & Keyboard Shortcuts

- **Customizable Shortcuts:** Added a keyboard icon in the POS header. Users can now view and remap keyboard shortcuts (e.g., F1 for Orders) and have them persist in the database.
 - **Visual Feedback:** Added amber border highlights and instructions for the shortcut rebinding process.
 - **Build Errors:** Fixed a lingering SWC build bug related to JSX ternary operators and duplicate tags in the shortcuts modal.
-

Generated by Antigravity